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NTI
National Telecommunication Institute

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Ministry of Communications and Information Technology

Digital Egypt Pioneers Initiative Fortinet Project
Implementing VPN Solutions with FortiGate

Presented By:

- Michael Boktor Abdullah – 21081516
- Mina Wafik Halim - 21004210
- Yasmin Reda Mahmoud - 21044508
- Salma Mohamad Gad – 21032453

Presented To:

- Dr. Hussein Harb
- ENG. Ahmed Mahmoud

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Agenda

- Introduction
- Week 1 Deliverables
- Week 2 Deliverables
- Week 3 Deliverables
- Conclusion
- Q&A

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Overview of VPNs and Their Importance

A Virtual Private Network (VPN) is a secure connection that allows users to access a private network over the internet. VPNs encrypt data traffic, ensuring confidentiality, integrity, and anonymity. They are crucial for businesses and individuals as they:

- Protect sensitive data from cyber threats.
- Enable secure remote access for employees.
- Ensure privacy when using public networks.
- Facilitate safe communication between branch offices or across geographies.

FORTINET

- A leading global provider of cybersecurity and networking solutions
- Established in 2000, trusted by enterprises, governments, and service providers
- Creator of FortiGate, one of the most widely deployed next-generation firewalls

Solution Areas

- Network security (NGFW, IPS, Web Filtering)
- Secure remote access & VPN technologies
- SD-WAN and branch connectivity
- Cloud and data center protection

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Why FortiGate for VPN Solutions

FortiGate, developed by Fortinet, is a leading network security platform that offers robust VPN solutions. It is widely chosen because it:

- Provides high-performance, hardware-accelerated VPN connections.
- Supports both SSL and IPsec VPNs for flexible deployment.
- Integrates advanced security features like firewall, intrusion prevention, and malware protection.
- Offers easy management through a centralized interface, making VPN deployment and monitoring efficient.

FORTINET

Project Stages

1- SSL VPN Configuration

- Remote access for users
- Authentication, portal settings, and access policies

2- IPsec Site-to-Site VPN

- Private communication between two networks
- Routing and traffic selectors to enable encrypted connectivity

3- SD-WAN Integration

- Link performance monitoring
- Intelligent VPN traffic steering for reliability and continuity

Week 1 – VPN Concepts & SSL VPN Configuration on FortiGate

What is a VPN?

A Virtual Private Network (VPN) is a technology that creates a secure, encrypted tunnel between a user and a network over the internet.

Why do we use a VPN?

- Protects data from interception
- Allows safe remote access to internal resources
- Connects networks securely over public channels
- Ensures privacy and confidentiality

Types of VPNs

Site-to-Site VPN (Network ↔ Network)

Connects entire networks (e.g., HQ ↔ branch) •

Used for permanent connections •

Ideal for businesses with multiple locations •

Runs mainly on IPsec •

Always-on and transparent to users •

Remote Access VPN (User ↔ Network)

Allows individual users to connect remotely •

Ideal for employees working from home/outside •

Two main types :

SSL VPN

IPsec Remote Access

Requires user authentication •

SSL VPN vs IPsec VPN

- **SSL VPN**

Works over HTTPS (TCP 443)

Uses SSL/TLS encryption

Easy to access (browser/FortiClient)

Works smoothly behind NAT & firewalls

Best for remote users / work from home

- **IPsec VPN**

Uses IPsec protocols •

Requires a VPN client •

More secure for managed corporate devices •

Very stable for site-to-site tunnels •

Preferred for permanent office connections •

Configuration (Week 1 – SSL VPN)

Step 1 – Interface Configuration:

- Port1 (WAN) was configured to receive a DHCP IP address for external connectivity.

- Port2 (LAN) was assigned the static subnet 192.168.100.1/24 to serve internal hosts.

Interfaces were verified to ensure proper WAN and LAN separation.

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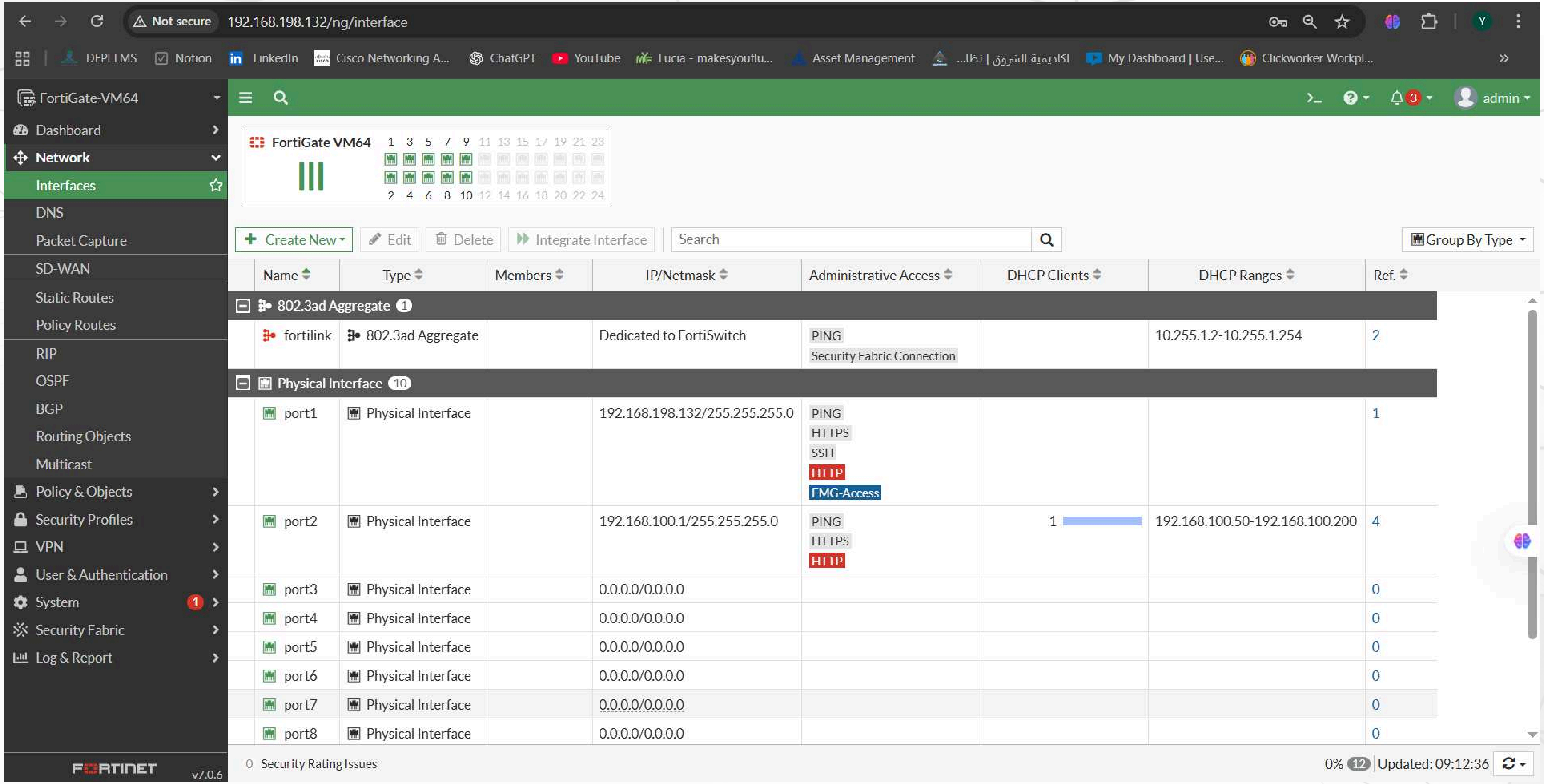

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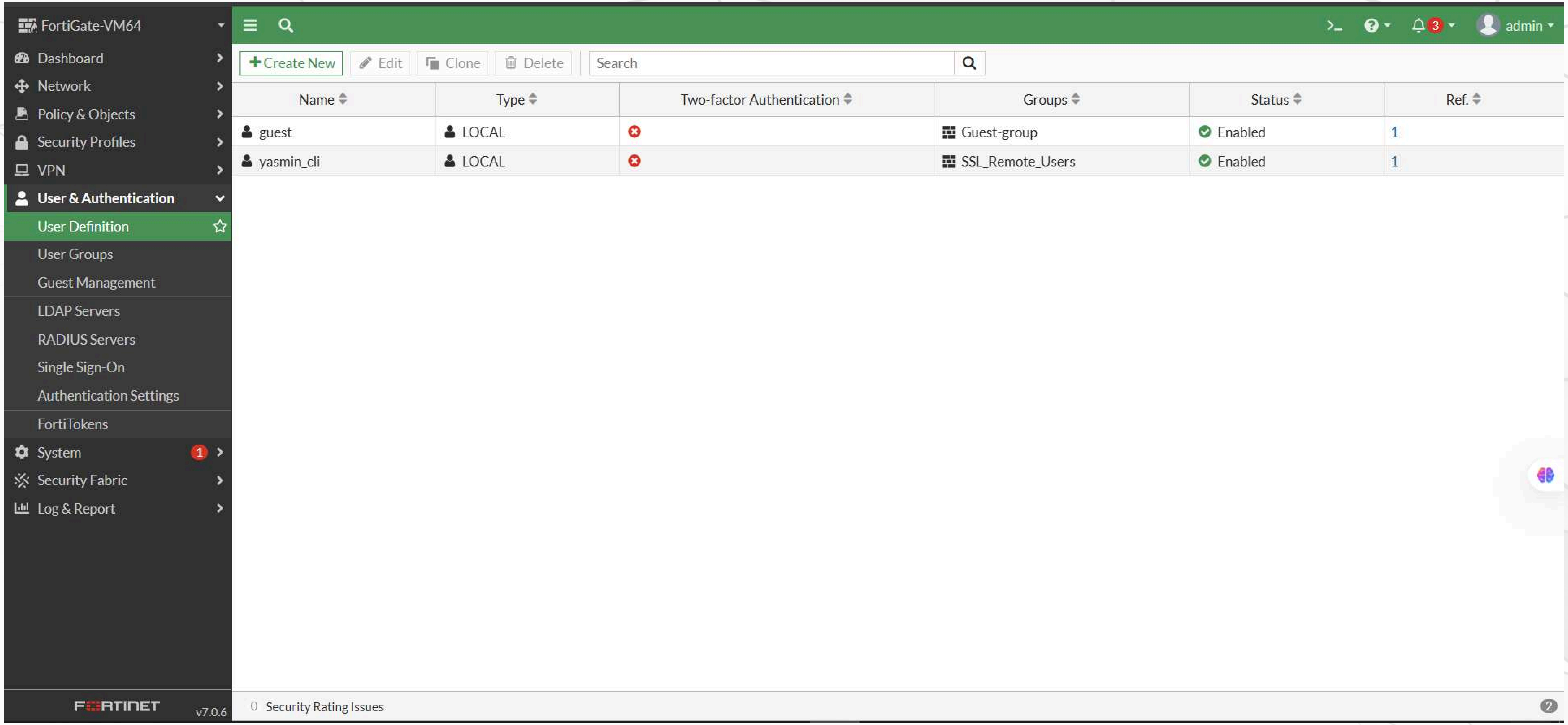
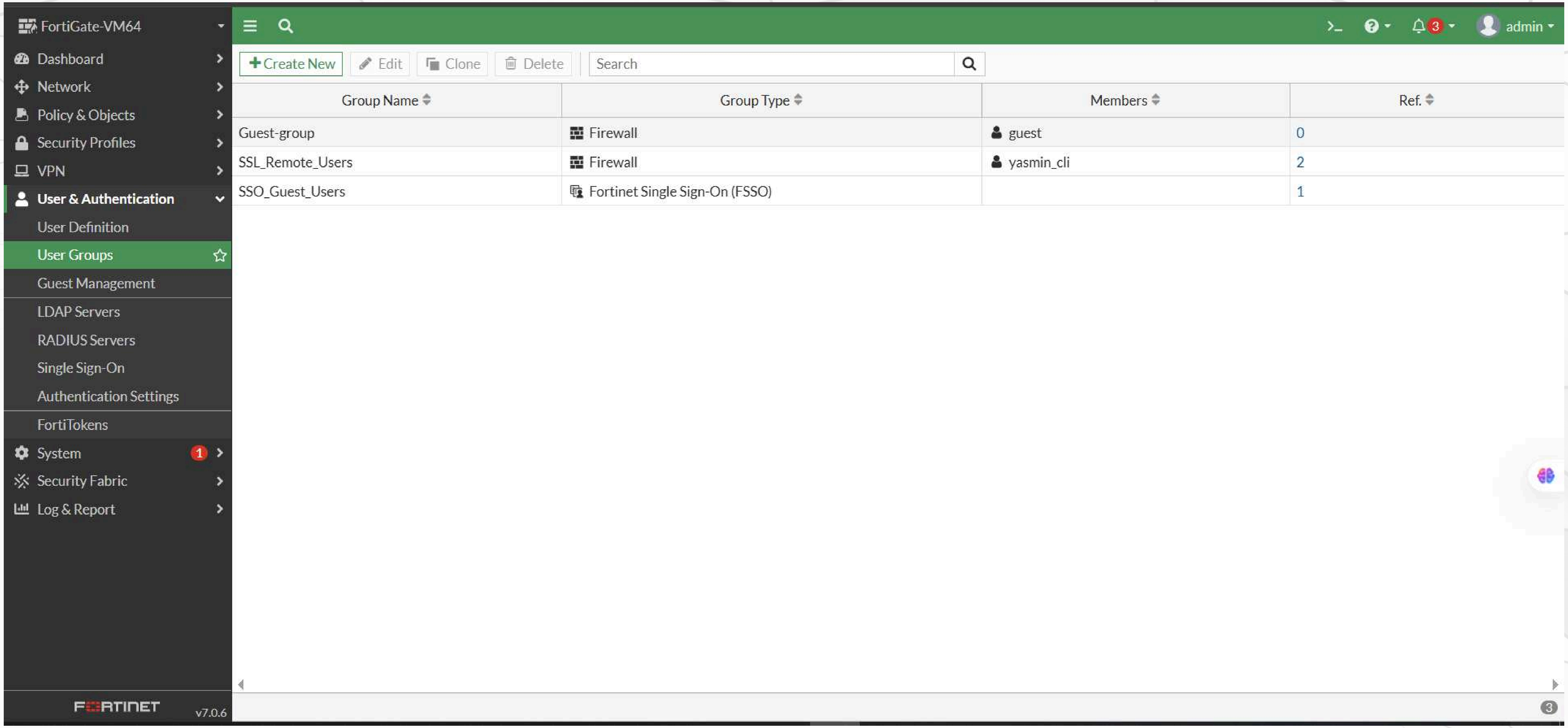
Step 2 – Local User Creation:

- A local user (yasmin_cli) was created for SSL-VPN authentication.
- The user was assigned a secure password and enabled for login.



Step 3 – User Group Configuration:

- A dedicated group named SSL_Remote_Users was created.
  - The local user yasmin_cli was added to this group.



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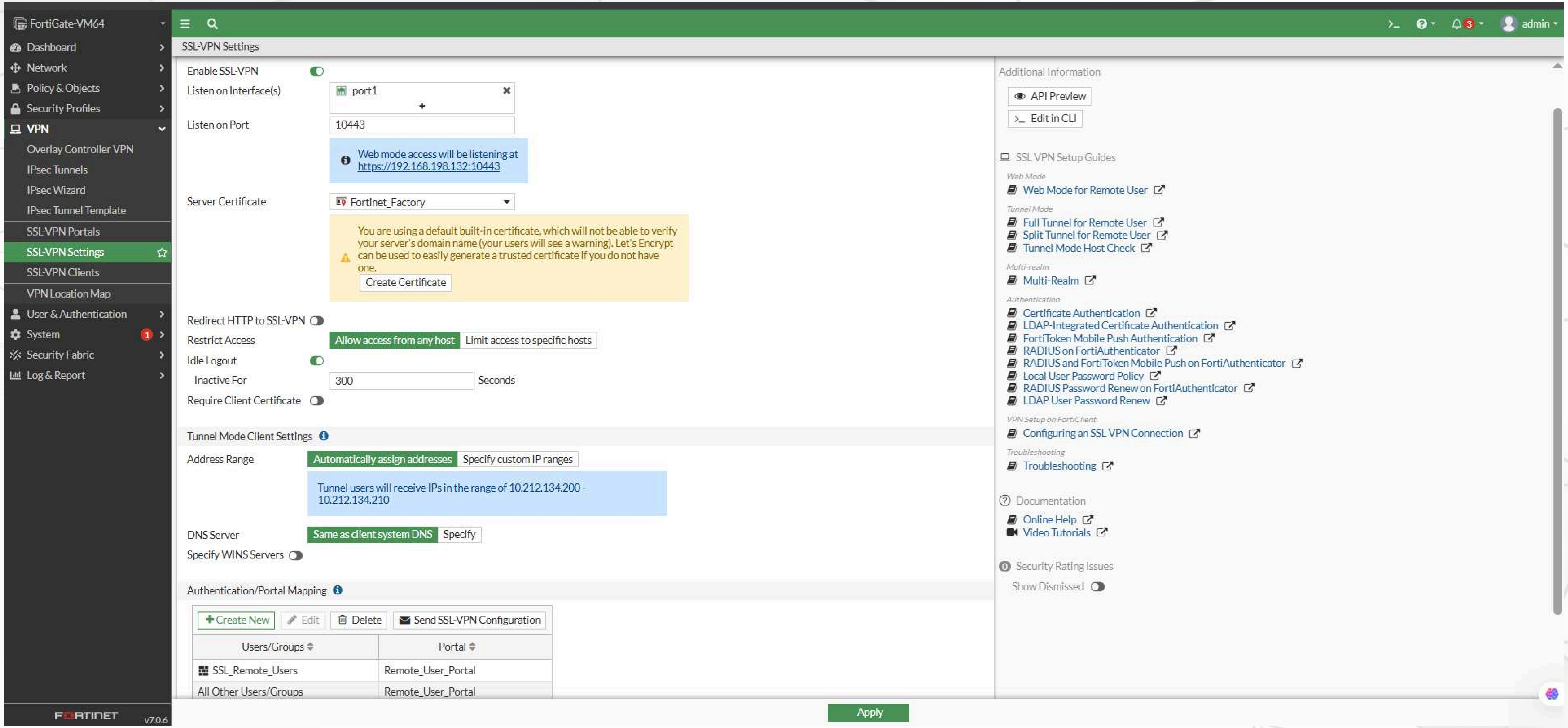
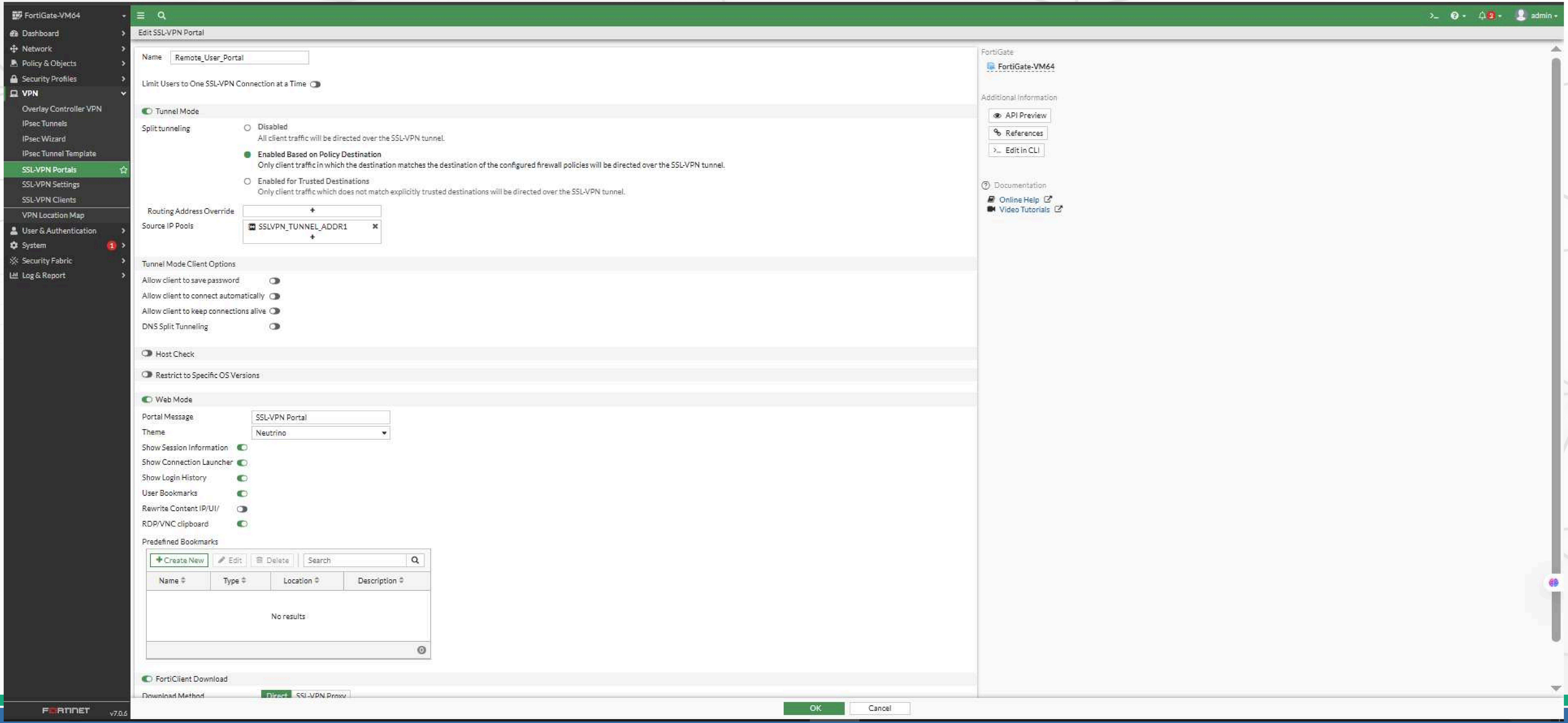
Step 4 – SSL VPN Settings:

- SSL-VPN was enabled on port1.



Step 5 – SSL-VPN Portal Setup:

- The Remote_User_Portal was configured.
- Split Tunneling was enabled to route internal traffic only.
- Routing override was set to the SSLVPN address pool.



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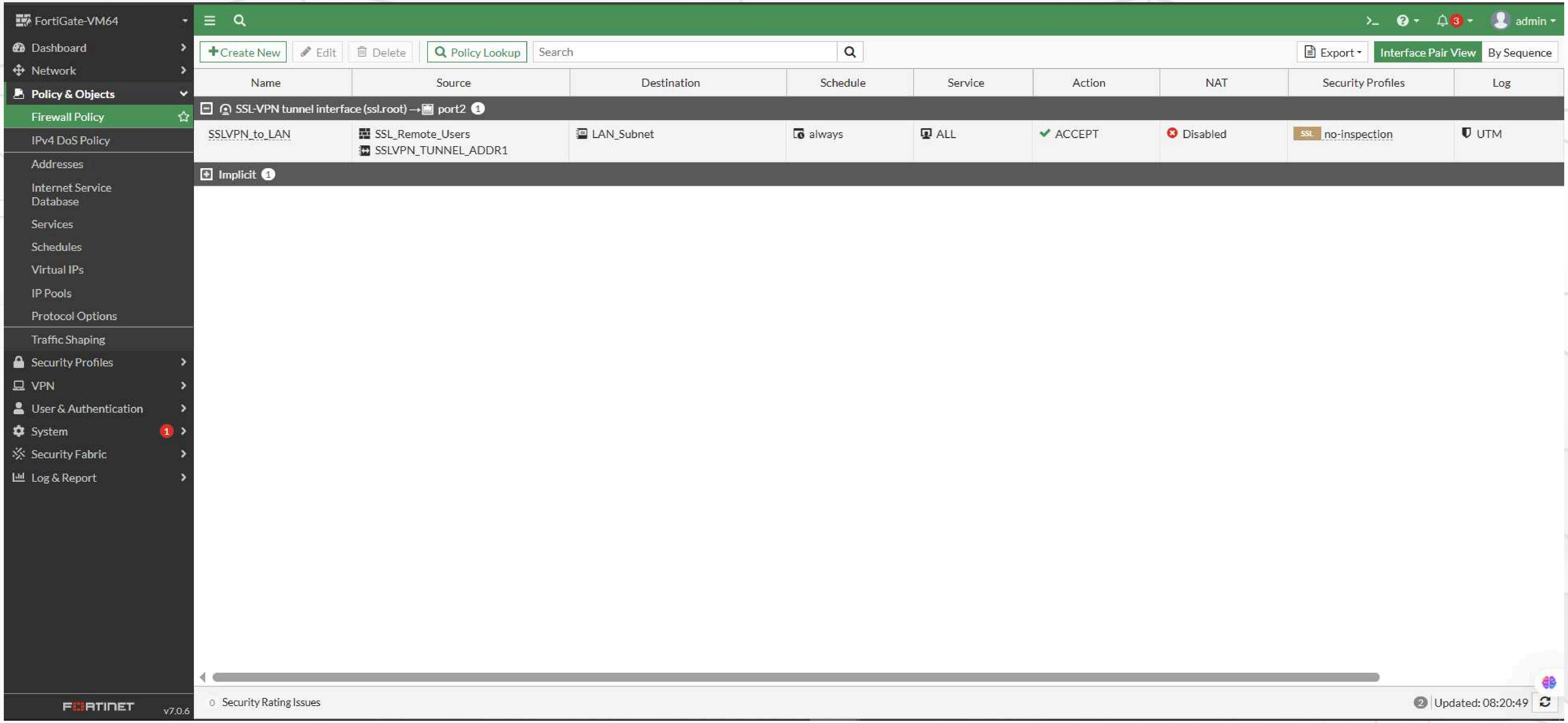
Step 6 – Firewall Policies:

- Policy created to allow SSLVPN → LAN user access to internal resources.



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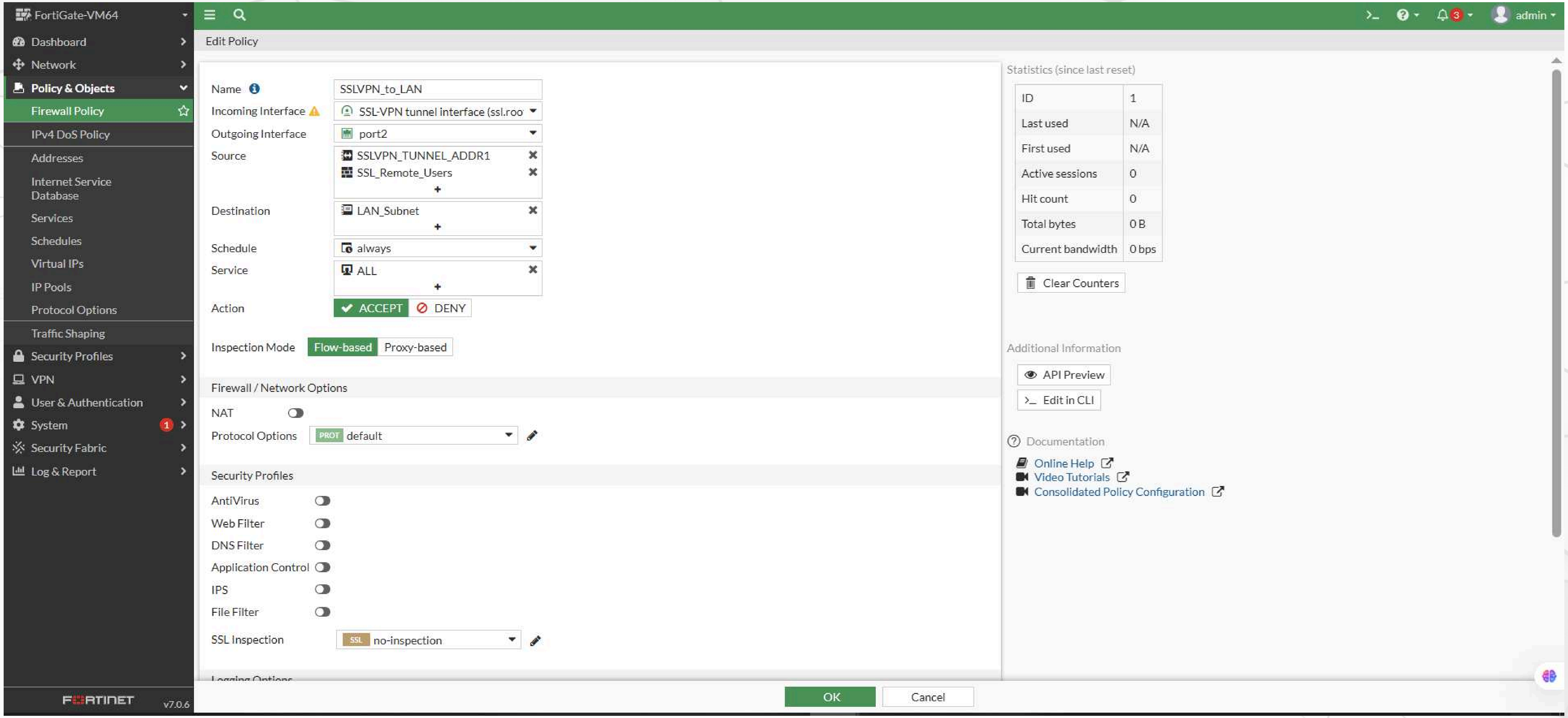

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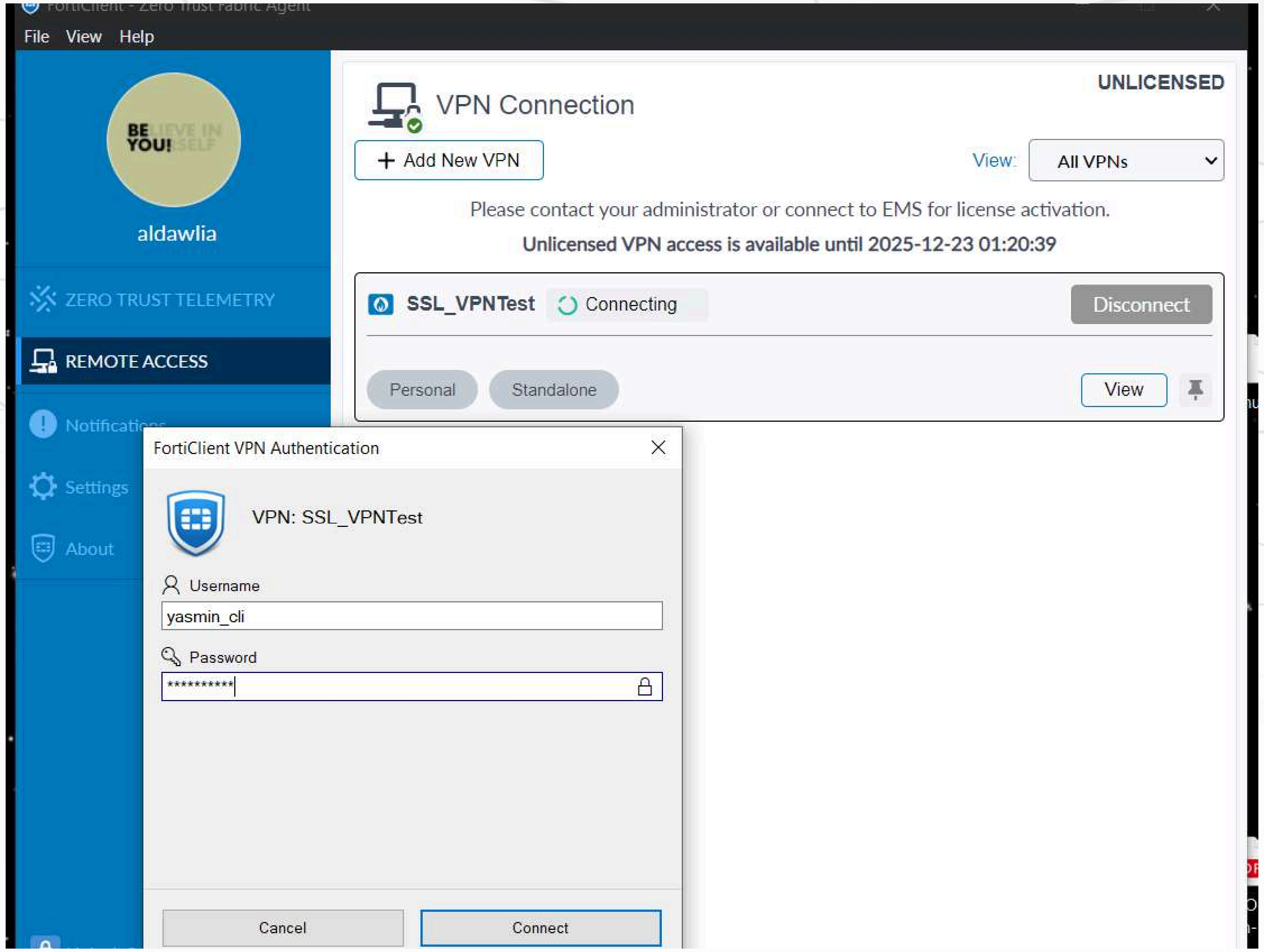
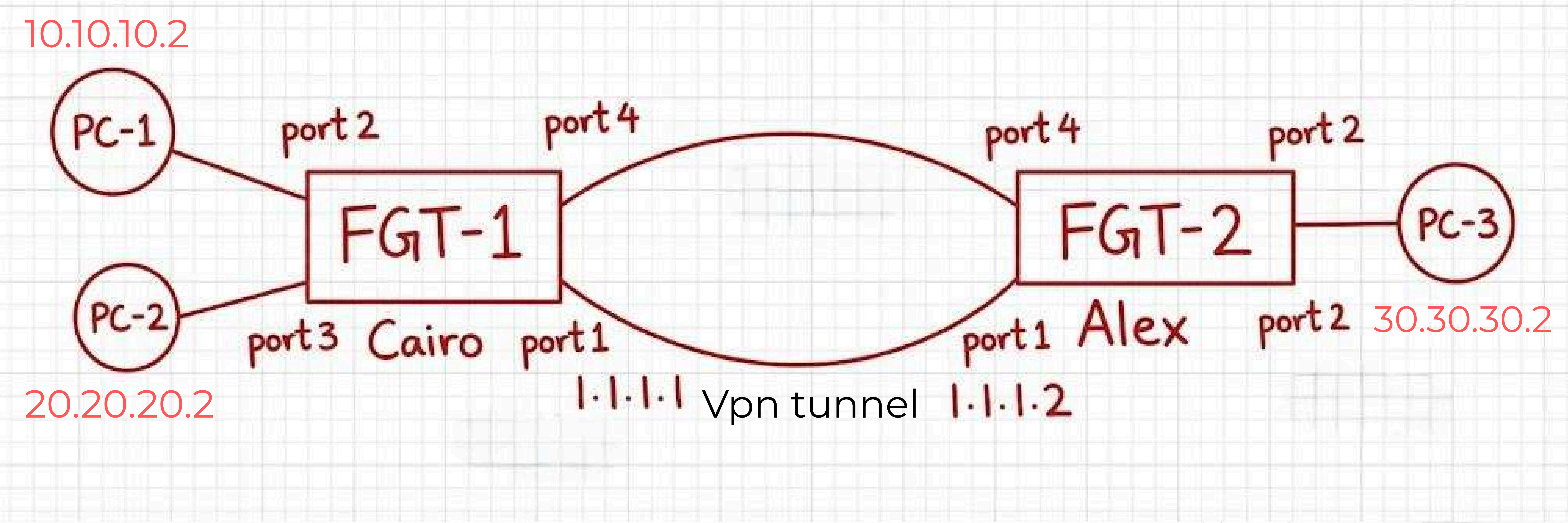
TEST



Week2: IPsec Site-to-Site VPN Tunnel Implementation

Topology Overview



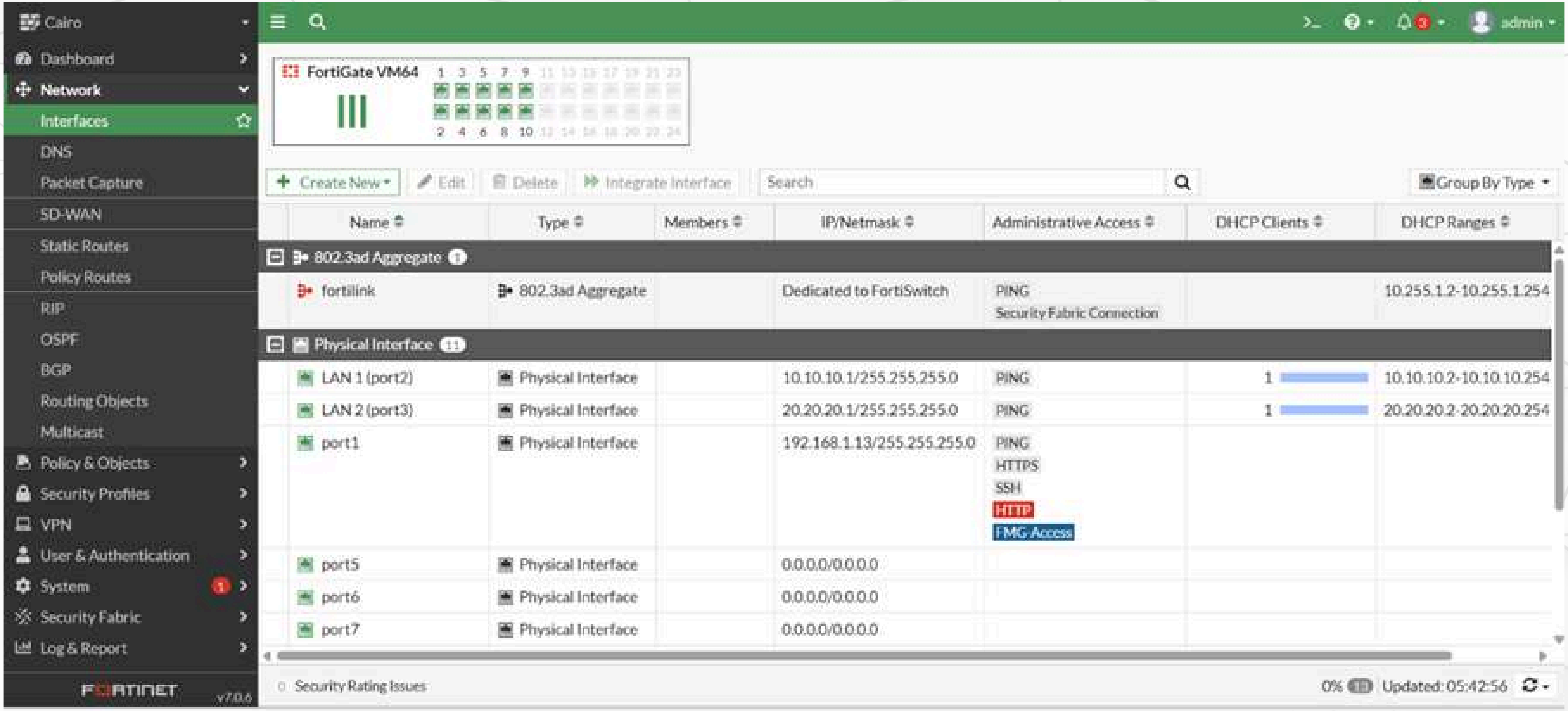
Goal Traffic:

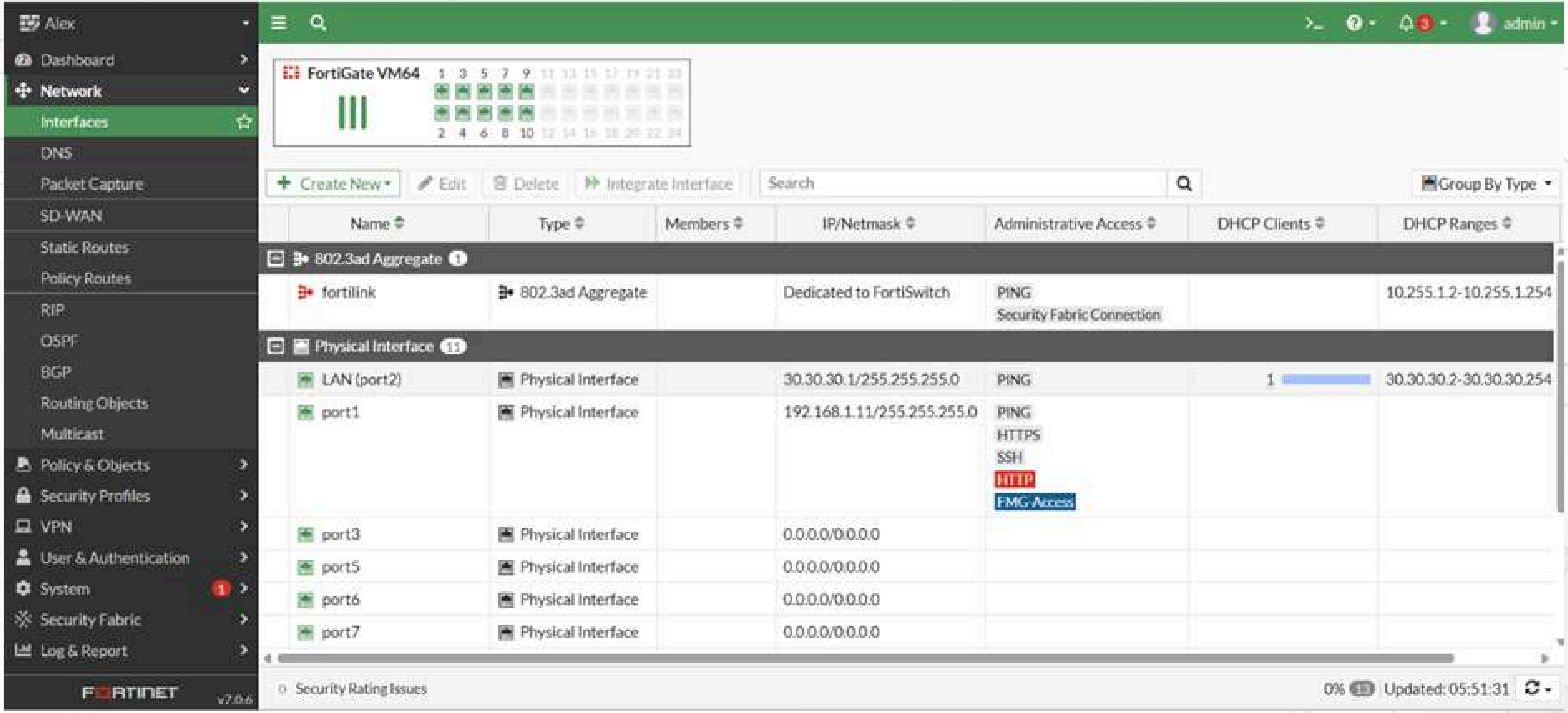
Secure communication primarily between the 10.10.10.0/24 and 30.30.30.0/24 networks.

Configuration

Step 1 – Interface Configuration:

- Port1 was configured as the management interface on both firewalls.
- IP addresses were assigned to all required interfaces to provide both internal and external connectivity.
- Each site was configured with its correct internal subnets and WAN interfaces.


Cairo
Dashboard
Network
Interfaces
DNS
Packet Capture
SD-WAN
Static Routes
Policy Routes
RIP
OSPF
BGP
Routing Objects
Multicast
Policy & Objects
Security Profiles
VPN
User & Authentication
System
Security Fabric
Log & Report
FORTINET v7.0.6
admin
FortiGate VM64
Create New
Edit
Delete
Integrate Interface
Search
Group By Type
Name
Type
Members
IP/Netmask
Administrative Access
DHCP Clients
DHCP Ranges
802.3ad Aggregate
fortilink
802.3ad Aggregate
Dedicated to FortiSwitch
PING
Security Fabric Connection
10.255.1.2-10.255.1.254
Physical Interface
LAN 1 (port2)
Physical Interface
10.10.10.1/255.255.255.0
PING
1
10.10.10.2-10.10.10.254
LAN 2 (port3)
Physical Interface
20.20.20.1/255.255.255.0
PING
1
20.20.20.2-20.20.20.254
port1
Physical Interface
192.168.1.13/255.255.255.0
PING
HTTPS
SSH
HTTP
FMG-Access
port5
Physical Interface
0.0.0.0/0.0.0.0
port6
Physical Interface
0.0.0.0/0.0.0.0
port7
Physical Interface
0.0.0.0/0.0.0.0
Security Rating Issues
Updated: 05:42:56


Alex
Dashboard
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Static Routes
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FortiGate VM64
Create New
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Members
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DHCP Clients
DHCP Ranges
802.3ad Aggregate
fortilink
802.3ad Aggregate
Dedicated to FortiSwitch
PING
Security Fabric Connection
10.255.1.2-10.255.1.254
Physical Interface
LAN (port2)
Physical Interface
30.30.30.1/255.255.255.0
PING
1
30.30.30.2-30.30.30.254
port1
Physical Interface
192.168.1.11/255.255.255.0
PING
HTTPS
SSH
HTTP
FMG-Access
port3
Physical Interface
0.0.0.0/0.0.0.0
port5
Physical Interface
0.0.0.0/0.0.0.0
port6
Physical Interface
0.0.0.0/0.0.0.0
port7
Physical Interface
0.0.0.0/0.0.0.0
Security Rating Issues
0%
Updated: 05:51:31

Configuration

Step 2 – VPN Tunnel Configuration

- An IPsec VPN tunnel was created on both firewalls.
- The traffic selectors were configured as follows:
  - Cairo: 10.10.10.0/24 → 30.30.30.0/24
  - Alex: 30.30.30.0/24 → 10.10.10.0/24
- Phase 1 and Phase 2 parameters were matched to ensure successful negotiation of the VPN tunnel.

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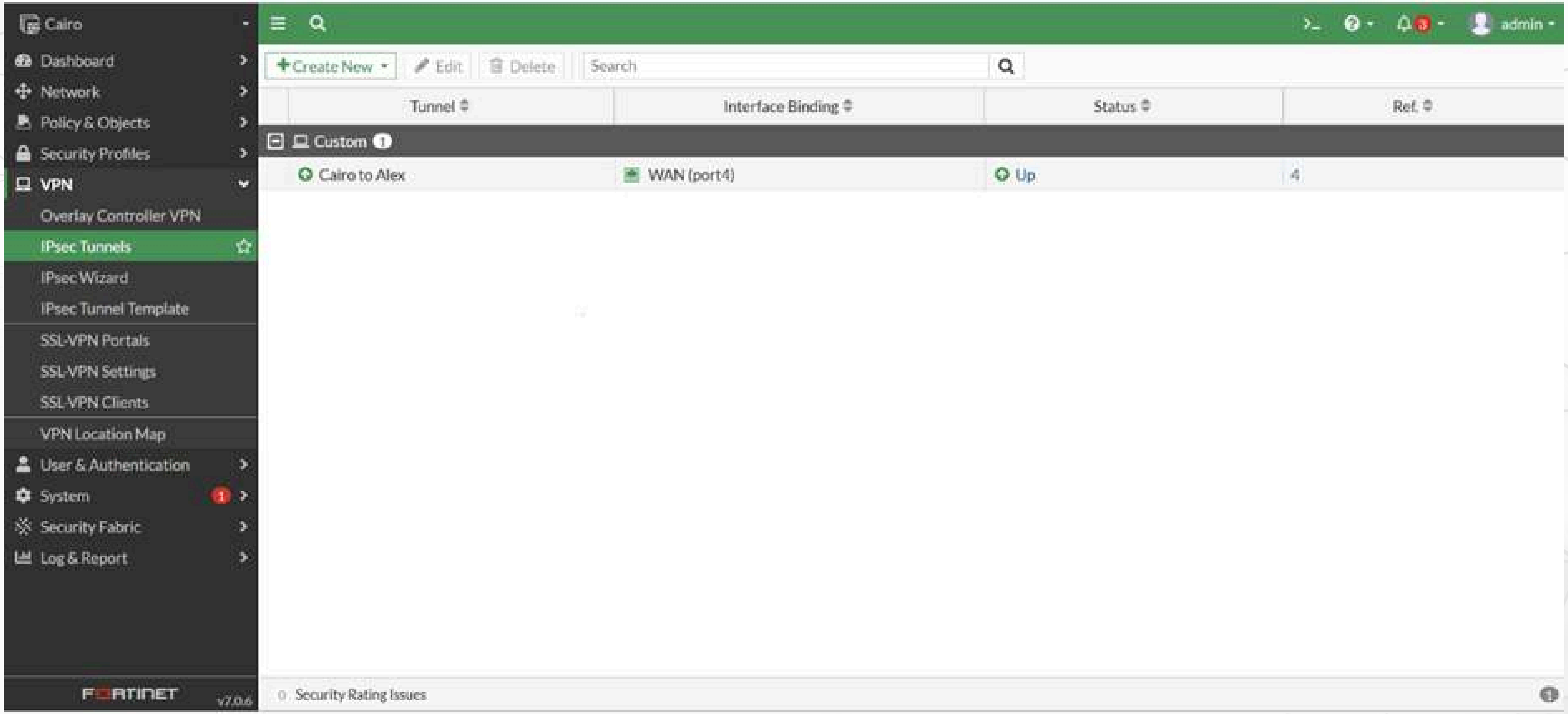

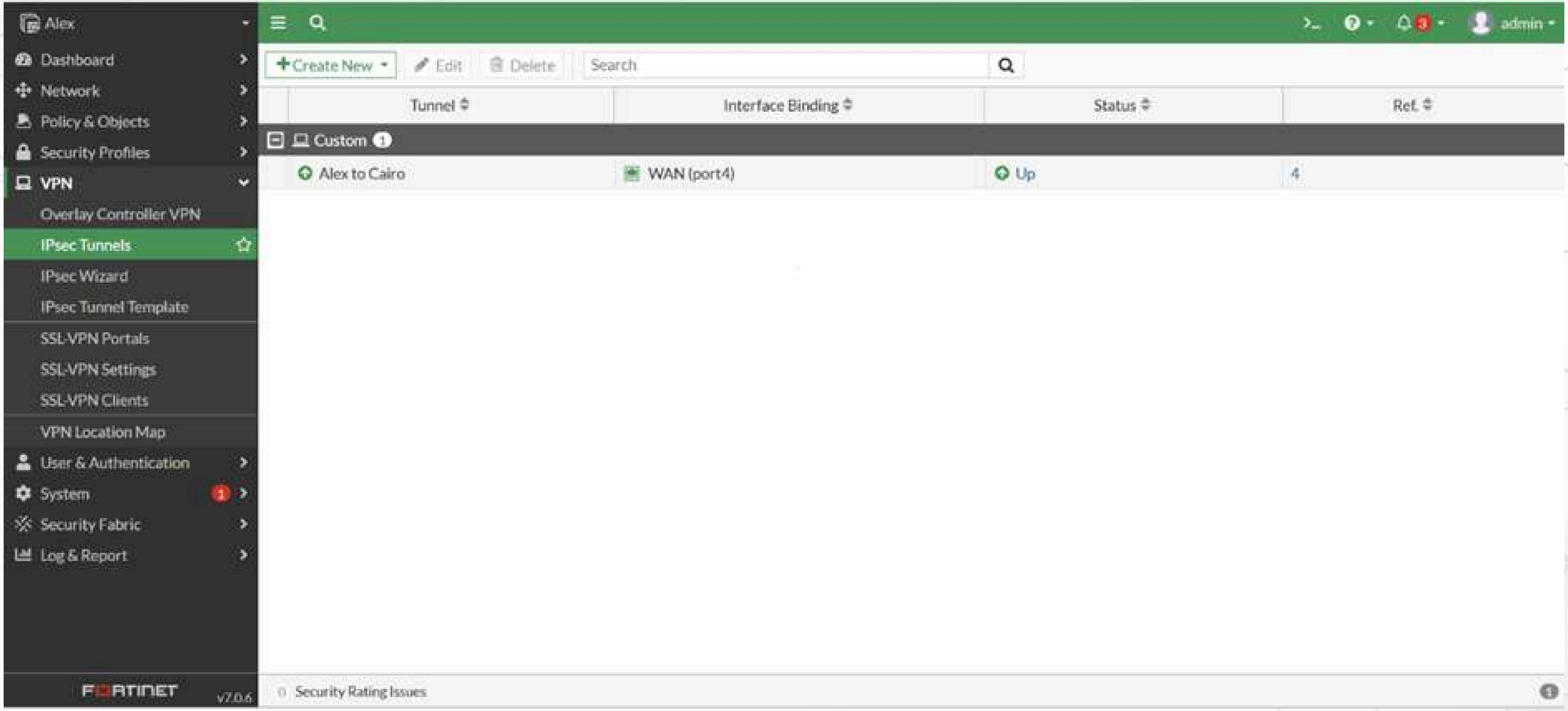
Alex
Dashboard
Network
Policy & Objects
Security Profiles
VPN
Overlay Controller VPN
IPsec Tunnels
IPsec Wizard
IPsec Tunnel Template
SSL-VPN Portals
SSL-VPN Settings
SSL-VPN Clients
VPN Location Map
User & Authentication
System
Security Fabric
Log & Report
FORTINET
v7.0.6
admin
Create New
Edit
Delete
Search
Tunnel
Interface Binding
Status
Ref.
Custom
Alex to Cairo
WAN (port4)
Up
4
Security Rating Issues

Configuration

Step 3 – Static Route Configuration

- Static routes were added on each firewall so that traffic destined for the remote network is forwarded into the VPN tunnel.
- This ensures correct routing between the Cairo and Alex networks.

Step 4 – Firewall Policy Configuration

- Security policies were created to allow traffic between the internal networks and the VPN tunnel interface.
- Only traffic that matches the defined traffic selector is allowed through the tunnel.

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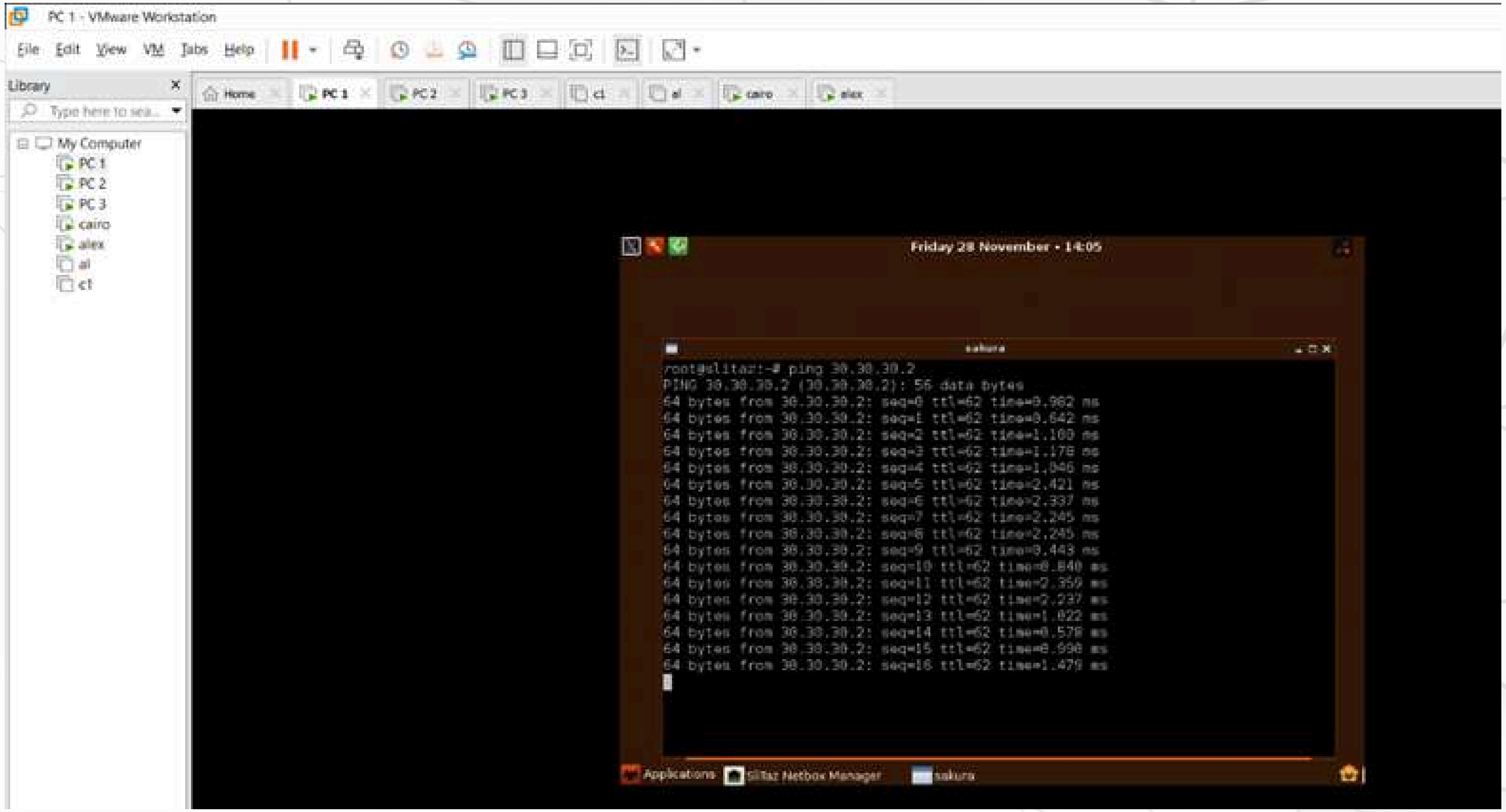
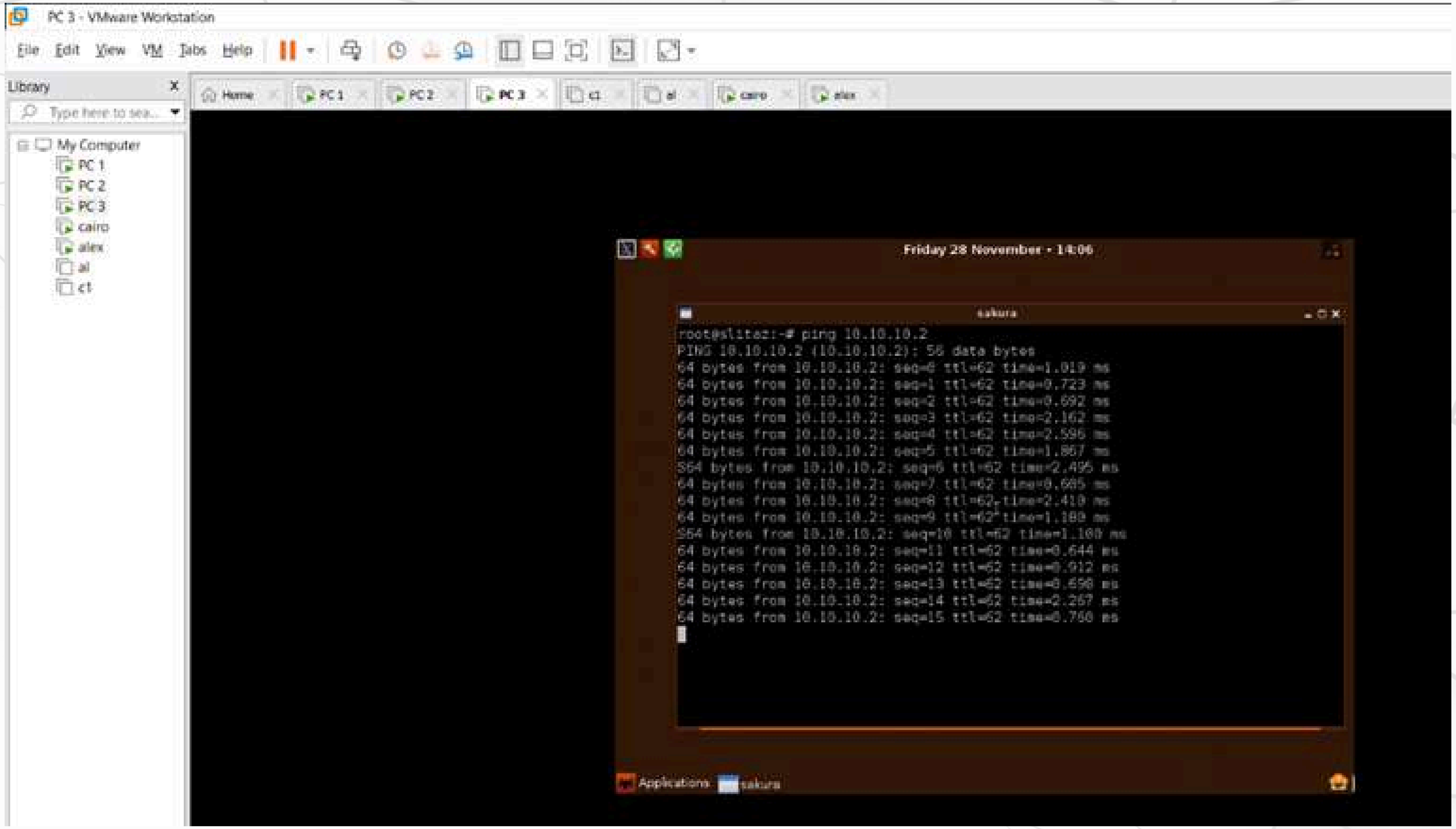
Connectivity Test Results

Ping Test: PC-1 to PC-3



Connectivity Test Results

Ping Test: PC-3 to PC-1



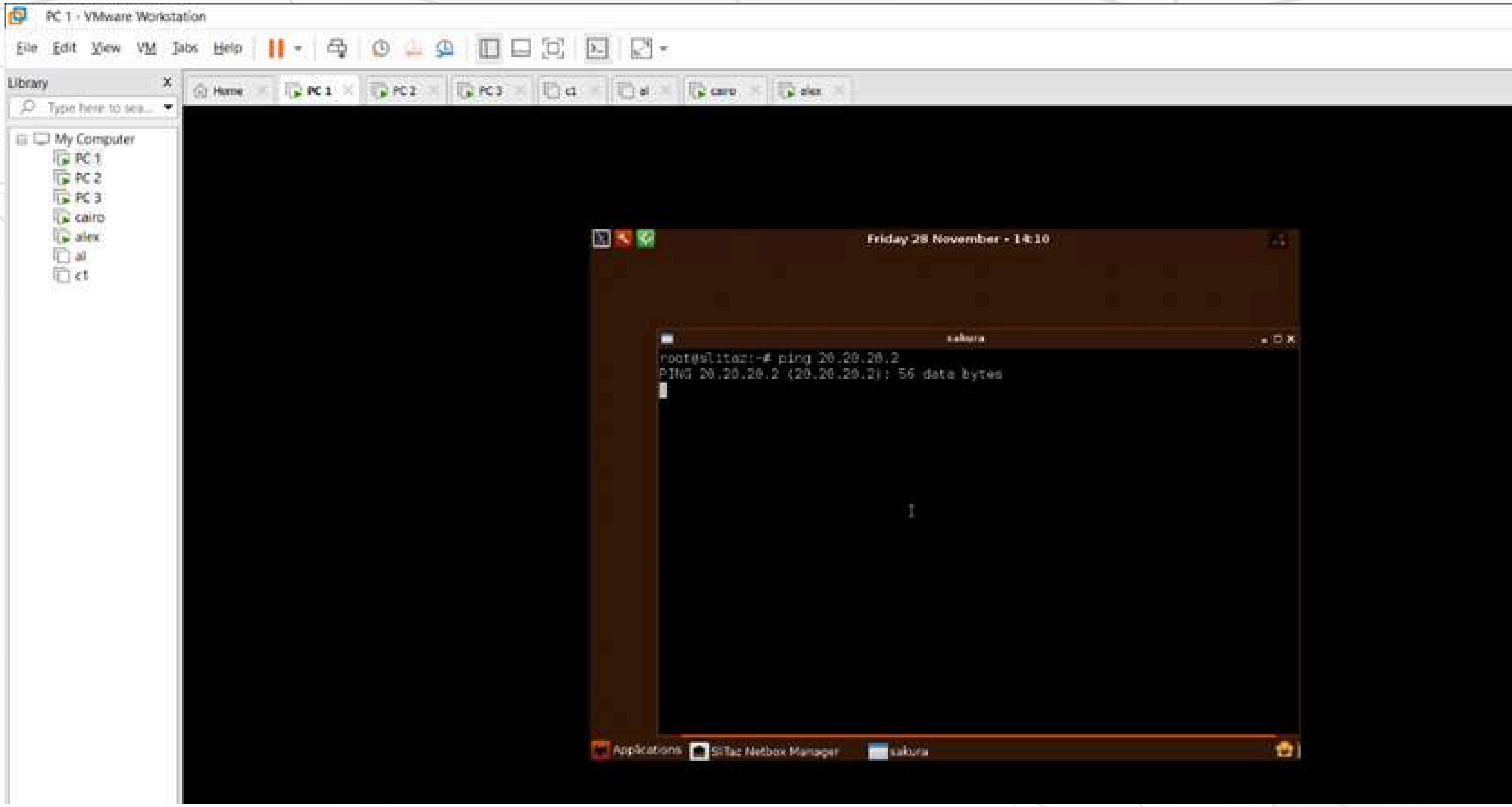
Connectivity Test Results

Comment:

- The ping was successful, confirming that the IPsec VPN tunnel is working correctly.
- Connectivity is allowed because the traffic selector includes 10.10.10.0/24 ↔ 30.30.30.0/24.

Connectivity Test Results

Ping Test: PC-1 to PC-2



Connectivity Test Results

Comment:

- The ping failed, which is the expected result.
- No traffic selector is configured between the 10 network and the 20 network, so the firewall drops this traffic.

Week 3: FortiGate VPN with SD-WAN

Configuration steps of SD-WAN on FortiGate

Figure 1:configure the ports to a wan

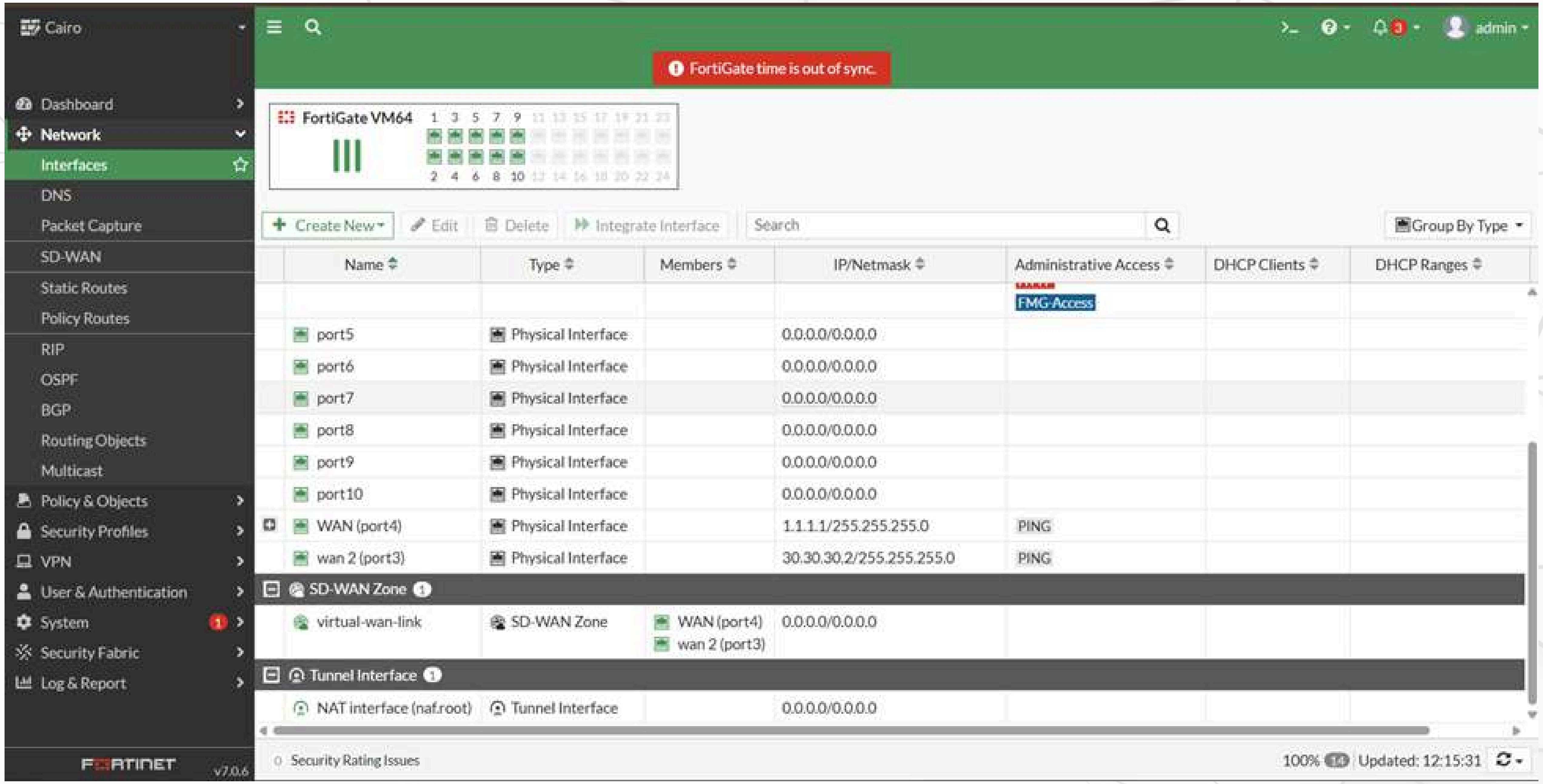


Figure 2 : create new SD-WAN member

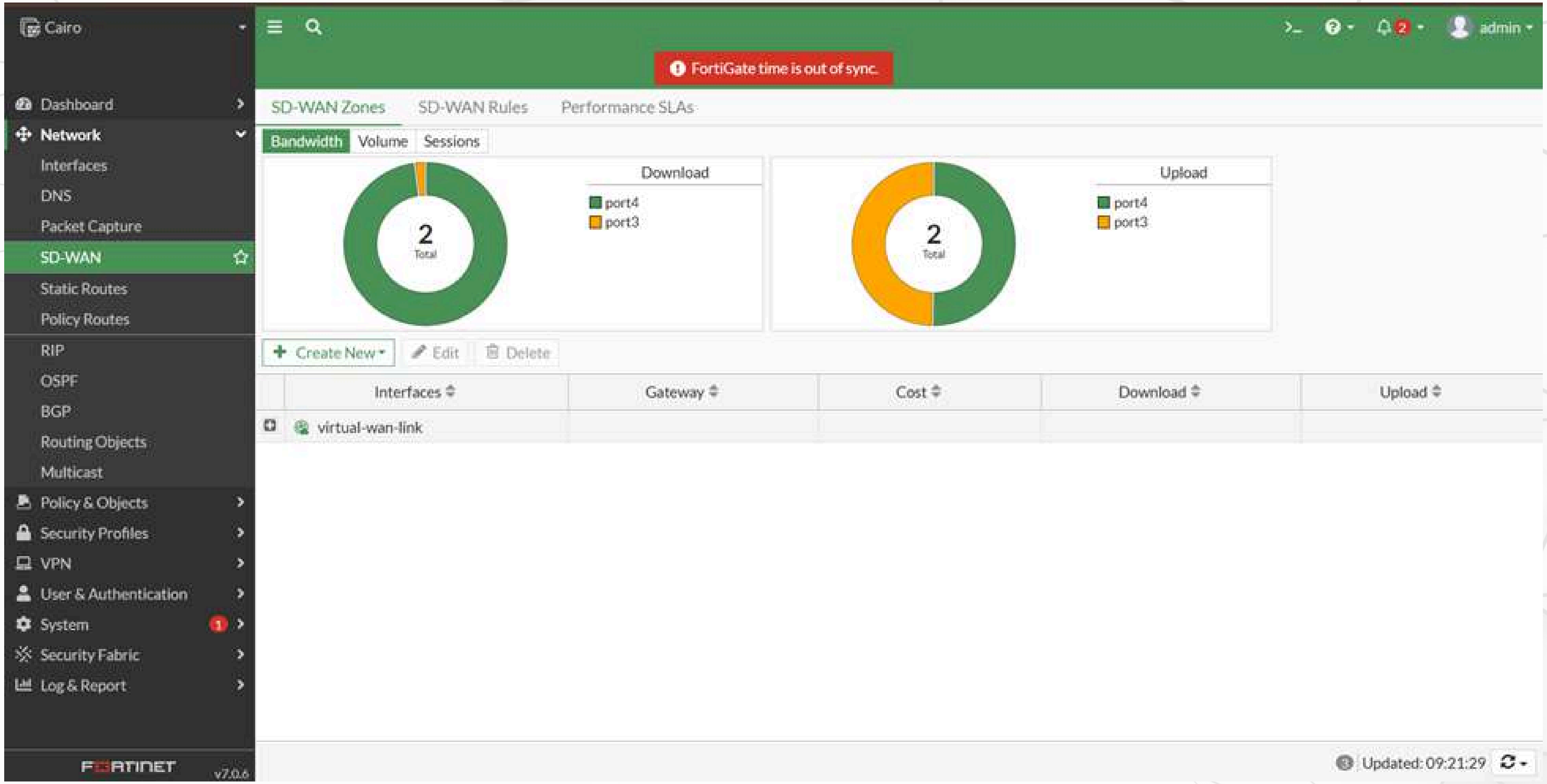


Figure 3: Define Link Performance Metrics (Performance SLA)

Purpose: This is the core intelligence. You define the acceptable quality targets (latency, jitter, loss) for the link carrying the VPN traffic. The FortiGate constantly checks both wan1 and wan2 against these targets.

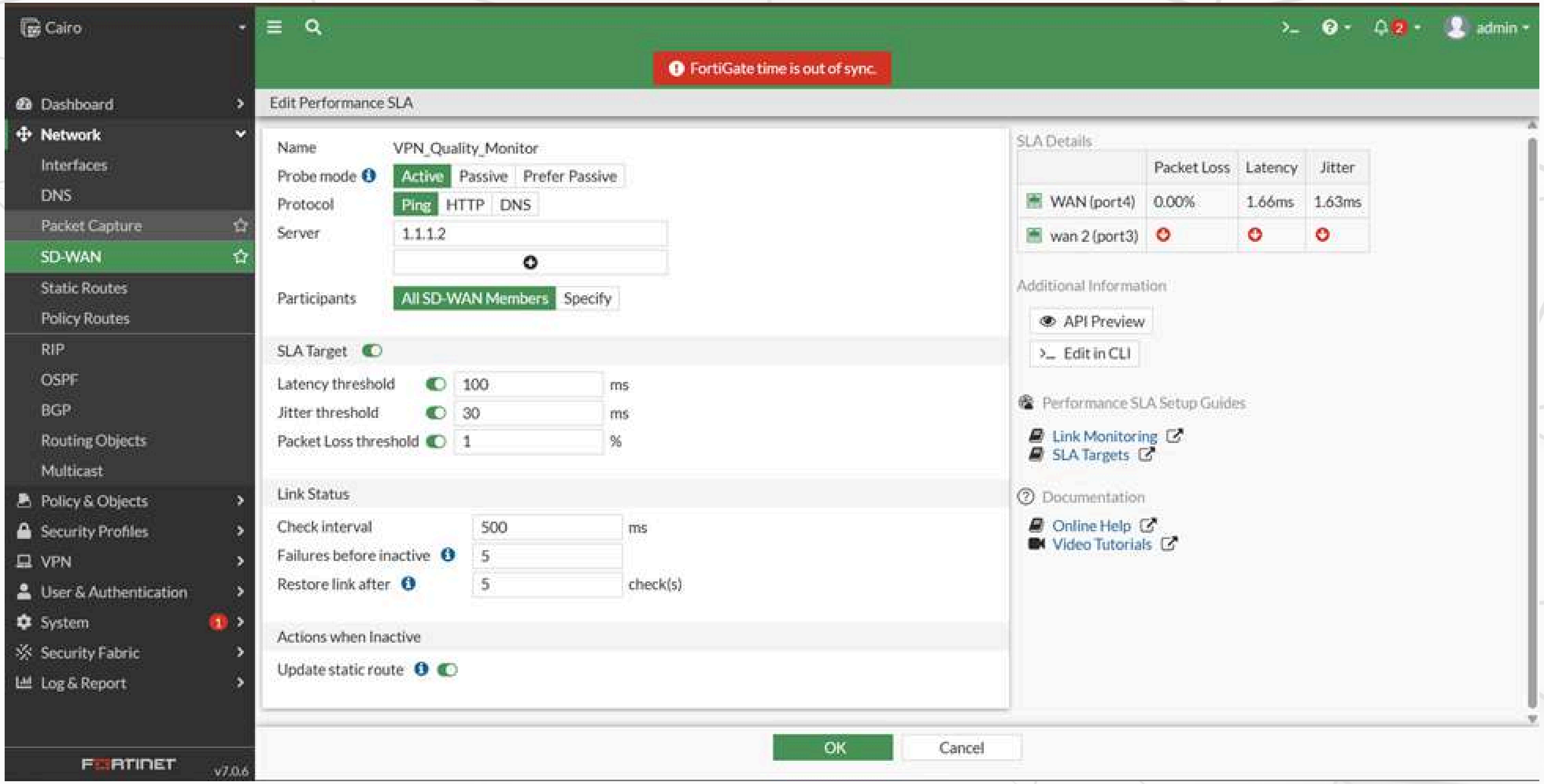


Figure 4 : Create the SD-WAN Rule (Traffic Steering Logic)

Purpose: This rule specifically directs the VPN traffic to use the best performing link based on the real-time data collected

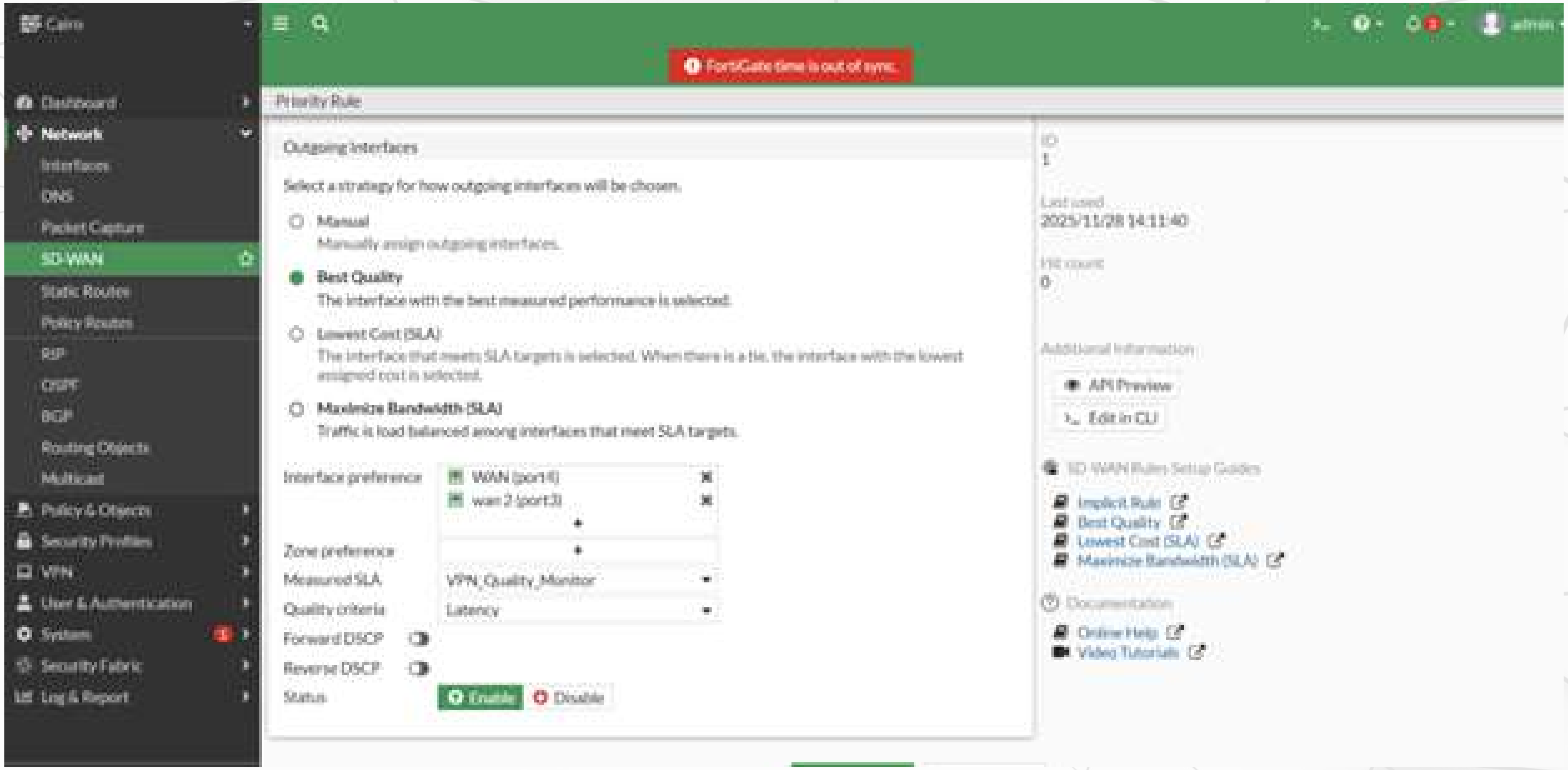
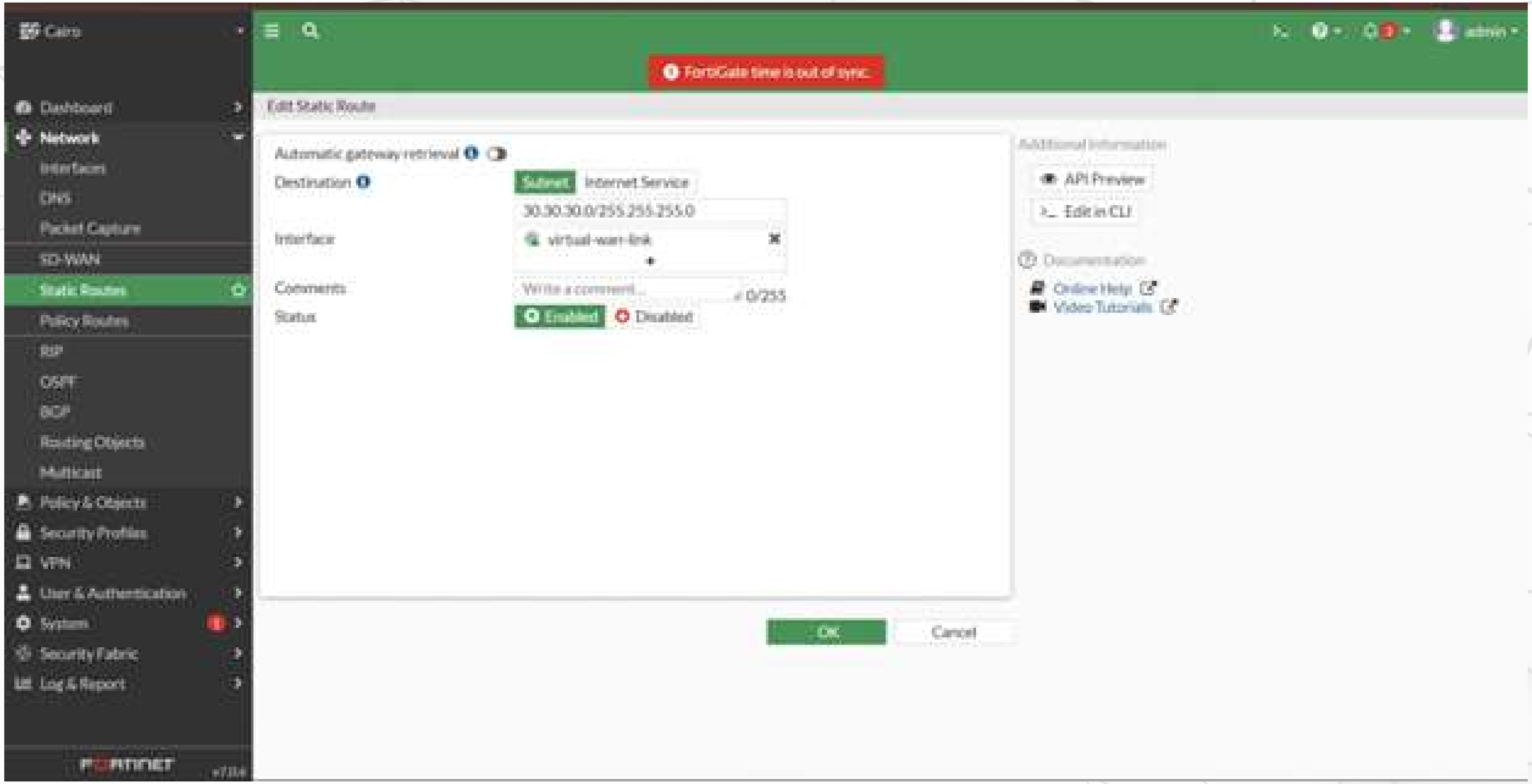


Figure 5 : Update the Static Route

Purpose: To make the SD-WAN rule effective, you must redirect the traffic bound for the remote VPN network away from a single physical interface and hand it over to the SD-WAN engine.



Conclusion

- Successfully implemented three FortiGate VPN solutions: **SSL, IPsec, and SD-WAN**.
- **SSL VPN** → Delivered secure remote access with user authentication and controlled permissions.
- **IPsec VPN** → Enabled encrypted site-to-site communication and verified tunnel stability through testing.
- **SD-WAN integration** → Improved VPN reliability by monitoring link metrics and automatically selecting optimal paths.
- Demonstrated real **configuration, troubleshooting**, and **validation**, not just theoretical concepts.

Any Questions ?

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